

In-Class Exercises

Writing Classes #2

Date Class

Write a class that represents a date. A date contains a year, a month, and a day, and has the following methods:

1. A Constructor that takes the month, day, and year as parameters, and sets them to instance variables.
2. A toString() method that returns the date in the format mm/dd/yyyy
3. A method called isSameDecade() that takes another Date as a parameter, and returns true if the dates are in the same decade, and false otherwise. For example, 1987 and 1981 are the same decade.

Actor Class

Write a class that represents an actor. The actor class stores the first name and last name. An actor has the following:

1. A constructor that takes the first name and last name as parameters, and sets them to instance variables.
2. An instance variable for `Movie[]`.
 - The `Movie` class will be written next!
3. An **`addMovie`** method that uses a (Using a `Movie` Class created in the next class) method to the Actor, (takes a `Movie` as a parameter)
4. A method **`getFullName()`** that returns the full name (i.e. first name and last name with a space between them) as a `String`.
5. **`toString()`** method that prints the Actor's name, followed by all of the `Movies` for the Actor.
 - Just call the `toString()` method from `Movie` class. You don't have to write this method yet.

Movie class

Write a class that represents a Movie. A movie has a title, an array of Actor, and a releaseDate. Use the Date class for the releaseDate, and an Actor class being written by another group for the actors. The class should have:

1. A **constructor** that takes the **title** and the **releaseDate** as parameters, and sets them to instance variables. The constructor should also create the array of actors but make it empty.
2. A method called **addActor** to the movie that takes an Actor as a parameter
3. A **toString()** method that returns the **Movie name** and **releaseDate** concatenated together.

Movie Theatre

1. Write a class that represents a Movie Theatre. A theatre has a Name, and an array of Movie).
 - a. A constructor that takes the Theatre Name and sets it to an instance variable, and also sets the array of Movie to a large empty array.
 - b. A method called addMovie() that takes a Movie as a parameter, and adds a movie to the theatre.
 - c. A toString() method that returns a String containing all the Movies (just use the toString() method of the Movie class when printing the movie)